33 kV FROM HV SWITCHBOARD
F00-SW-010
2000-F00-SLD-EL-10002

F21-SR-001
415V CIL4 SWITCHROOM

F21-TR-006
TRANSFORMER ROOM

F21-TR-006
33/0.433kV
2500kVA
Dyn11
z=6%
ANAF

SWR EARTH BAR
Tx EARTH BAR
NOTE 3
4000/ 1A
2.5VA
5P20

1600A
4 POLE
G
3~
F21-GE-006
415V
1000kVA

F21-SW-006
CIL4 415V MOTOR CONTROL CENTRE No.1

MCC EARTH BAR
NEUTRAL BAR
NOTE 1
1/700A
4000/ 5A
5VA
CL 1
4000/ 5A
30VA
CL0.5
TEST
PM
20A
63A
2A
32A
00F-08
ACB
4000A
2A
240VAC SPRING
CHARGE SUPPLY
2A
63A
AFD
60G
50
50G
51
51G
24VDC CONTROL POWER SUPPLY
NOTE 7
100A
100A
SD
NOTE 6

1600/ 5A
5VA
CL 1
1600/ 1A
2.5VA
5P20
AFD
01F-8
ACB
1600A
ATS
240VAC SPRING
CHARGE SUPPLY
24VDC CONTROL POWER SUPPLY
NOTE 7

CONTINUED ON SHT 2
2000-F22-SLD-EL-10202

EMERGENCY BUS 415V 3NE~ 50Hz 4000A 65kA for 1sec

06F-01 250A 90A 250A 250A
02R-01 32A 70A 32A 32A
03R-01 250A 90A
INC
06F-09 160A 70A ECR
05F-05 630A 250A EOL
04F-05 630A 250A EOL
03F-05 630A 250A EOL
INC
05R-05 630A 250A EOL

E-SHUT
F21SR001-LCS
NOTE 12

69kW
F21-DB-001
CIL4
415V SWITCHROOM
L&SP DB
2000-F21-SCM-EL-10205

ISL
F21UPS001-ISL
240VAC UPS
MAINTENANCE
BYPASS SWITCH
2000-F21-SCM-EL-10206

10kW
F21-DB-201
CIL4
415V SWITCHROOM
240VAC UPS DB
2000-F21-TLD-EL-10054

UPS
10kW
F21-UPS-001
CIL4
415V SWITCHROOM
240VAC UPS
2000-F21-TLD-EL-10054

69kW
F21-DB-301
CIL4
L&SP DB
2000-F21-SCM-EL-10207

CR
54kW
F21-CR-441
CIL4 AREA
MAINTENANCE CRANE
2000-F21-SCM-EL-10213

RTD M 3~ 250kW
F21-AG-411
CIL4 TANK 1
AGITATOR
2000-F21-SCM-EL-10208

RTD M 3~ 250kW
F21-AG-412
CIL4 TANK 2
AGITATOR
2000-F21-SCM-EL-10209

RTD M 3~ 250kW
F21-AG-413
CIL4 TANK 3
AGITATOR
2000-F21-SCM-EL-10210

RTD M 3~ 250kW
F21-AG-414
CIL4 TANK 4
AGITATOR
2000-F21-SCM-EL-10211

NORTHERN STAR RESOURCES LIMITED

Approved	X
Approved as Noted	
Not Approved	
For Information Only	

Signed [redacted] Date 19/11/2025

ACCEPTANCE DOES NOT RELIEVE THE SUPPLIER / CONTRACTOR OF ITS RESPONSIBILITY OR LIABILITIES UNDER THE PURCHASE ORDER / CONTRACT

NOTES

1. EARTH-NEUTRAL LINK SHALL BE INSTALLED IN INCOMER TIER. SIGNAGE SHALL BE PLACED ON MCC, TRANSFORMER, GENERATOR INCOMER AND GENERATOR READING "M.E.N. LINK LOCATED IN F21-SW-006 INCOMER TIER". CABLE SHALL BE IDENTIFIED C/W LABEL STATING "M.E.N. LINK - DO NOT REMOVE".
2. SWITCHBOARD SHALL BE EXTENSIBLE AND BE PROVIDED WITH THE FACILITY TO EXTEND THE MAIN BUSBAR AND APPEND ADDITIONAL TIERS AT ONE END WITHOUT MAJOR STRUCTURAL DISASSEMBLY.
3. UPSTREAM TRANSFORMER FEEDER SHALL BE TRIPPED ON ACTIVATION OF SWITCHROOM EMERGENCY PUSHBUTTON, DETECTION OF EARTH FAULT OR ARC FAULT DETECTION.
4. SWITCHBOARD VENDOR SHALL SELECT SWITCHGEAR TO PROVIDE TYPE-2 CO-ORDINATION SUITABLE FOR HIGH EFFICIENCY MOTORS.
5. ALL DRIVE I/O VIA COMMS NETWORK TO DCS. FOR COMMUNICATIONS BLOCK DIAGRAM REFER 2000-F21-BLK-IC-10001.
6. SURGE DIVERTERS SHALL PROVIDE FEEDBACK TO DCS VIA VOLT-FREE CONTACTS. SURGE DIVERTER FUSES SHALL BE SIZED BY VENDOR TO SUIT DUTY.
7. MINIATURE CIRCUIT BREAKERS SHALL BE SIZED AND RATED TO SUIT ACB AND INCOMER REQUIREMENTS.
8. THERMAL OVERLOAD UNITS MAY BE SUBSTITUTED WITH ELECTRONIC (MAGNETIC) OVERLOAD UNITS TO SUIT DUTY.
9. TOLs TO BE MOUNTED IN PURPOSE-BUILT ENCLOSURE BENEATH VSD. TOLs MAY BE INCLUDED IN MCC MODULE WITH ENGINEERING APPROVAL.
10. ANY DRIVE RTDs SHALL BE WIRED TO DEDICATED REMOTE I/O CARDS IN THE FIELD. RTD RELAYS MAY BE INCLUDED IN THE MCC MODULE WITH ENGINEERING APPROVAL.
11. CIRCUIT BREAKER TO BE SUBSTITUED FOR FUSED-SWITCH WHERE REQUIRED BY EQUIPMENT MANUFACTURERS GUIDELINES.
12. EMERGENCY SHUTDOWN (E-SHUT) PUSH BUTTON N/O WIRED TO SHUNT TRIP NOMINATED CIRCUIT BREAKERS.

LEGEND

- DIRECT ON-LINE MOTOR STARTER GENERIC - NOTE 4, 5
- DIRECT ON-LINE MOTOR STARTER REVERSING - NOTE 4, 5
- EOL — ELECTRONIC OVERLOAD UNIT NOTE 4, 5
- THERMAL OVERLOAD UNIT NOTE 8
- VARIABLE SPEED DRIVE NOTE 4, 5
- EARTH LEAKAGE RELAY NOTE 4, 5
- FIELD-MOUNTED DECONTACTOR
- MOTOR THERMISTOR (ONLY REQ. FOR MOTORS >37kW - 250kW)
- RTD — MOTOR RTD (ONLY REQ. FOR MOTORS >250kW)
- MOTOR HEATERS (ONLY REQ. FOR MOTORS >90kW)
- 49 63 71 — TRANSFORMER PROTECTIVE DEVICES (REFER ANSI STANDARDS FOR DESCRIPTIONS)
- CIRCUIT BREAKER (NOTE 12)
- xxxA ST TRIP SET POINT / xxxA CB SIZE
- AFD — ARC FLASH DETECTION
- ECR — EARTH CONTINUITY RELAY
- 1 E-SHUT — SWITCHROOM EMERGENCY SHUTDOWN
- 63A 3 PHASE SOCKET OUTLET

DRG No.	REFERENCES
2000-F21-SCM-EL-10204	F21-SW-006 415V MCC ARC FLASH SYSTEM SCHEMATIC DIAGRAM
2000-F21-SCM-EL-10203	F21-SW-006 415V MCC 24VDC DISTRIBUTION SCHEMATIC DIAGRAM
2000-F21-SCM-EL-10201	F21-SW-006 415V MCC INCOMER SCHEMATIC DIAGRAM
2000-F21-DRG-EL-10101	F21-SR-001 415V SWITCHROOM ELECTRICAL EQUIPMENT LAYOUT
2000-F21-BLK-IC-10001	F21-SR-001 415V SWITCHROOM AND REMOTE I/O COMMUNICATIONS BLOCK DIAGRAM
2000-F21-SLD-EL-10202	F21-SW-006 415V MCC SINGLE LINE DIAGRAM SHEET 2
2000-F00-SLD-EL-10002	F00-SW-010 33kV SWITCHBOARD SINGLE LINE DIAGRAM SHEET 1 OF 4
2000-F00-STD-EL-10001	GENERAL NOTES AND STANDARD DRAWINGS INDEX

REV	DATE	DESCRIPTION	DRAW	CHECK	DESIGN	TECH APP	PROJ APP
1	20OCT25	FAT REDLINES INCORPORATED	[redacted]				
0	28JAN25	APPROVED FOR CONSTRUCTION					

NORTHERN STAR RESOURCES LIMITED

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PROJECT: NORTHERN STAR RESOURCES LTD
KCGM GROWTH PROJECT

TITLE: FLOTATION TAILINGS CIL4
F21-SW-006 415V MCC
SINGLE LINE DIAGRAM SHEET 1

SHEET	SCALE	DRG No.	REVISION
A1	NONE	2000-F21-SLD-EL-10201	1

O:\CAD PROJECTS\23S NORTHERN STAR RESOURCES LTD\23517 KCGM GROWTH PROJECT\03 DELIVERABLES\EL\F21 - FLOTATION TAILINGS CIL 4\2000-F21-SLD-EL-10201DWG 06NOV25 11:07AM HOTSEAT04